

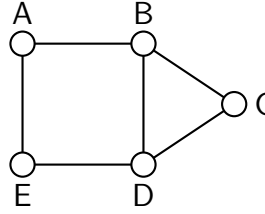
Indian Statistical Institute

BSDS II

Stochastic Process

Class Test II (sample solution)

1. Consider a random walk on the following graph:



- Write the transition matrix.
- Find an N , such that $p_{ii}^{(n)} > 0$ for any $i \in \{A, B, C, D, E\}$ and $n \geq N$.

[Note, we are not looking for minimum value of N .]

Sample Solution: The transition matrix of the random walk on the given graph is given by

$$S = \begin{matrix} & \begin{matrix} A & B & C & D & E \end{matrix} \\ \begin{matrix} A \\ B \\ C \\ D \\ E \end{matrix} & \begin{bmatrix} 0 & 1/2 & 0 & 0 & 1/2 \\ 1/3 & 0 & 1/3 & 1/3 & 0 \\ 0 & 1/2 & 0 & 1/2 & 0 \\ 0 & 1/3 & 1/3 & 0 & 1/3 \\ 1/2 & 0 & 0 & 1/2 & 0 \end{bmatrix} \end{matrix}.$$

First we observe that for $p_{AA}^{(2)} \geq p_{AB}p_{BA} > 0$, $p_{BB}^{(2)} \geq p_{BC}p_{CB} > 0$, $p_{CC}^{(2)} \geq p_{CD}p_{DC} > 0$, $p_{DD}^{(2)} \geq p_{DE}p_{ED} > 0$ and $p_{EE}^{(2)} \geq p_{EA}p_{AE} > 0$. Also, $p_{BB}^{(3)} \geq p_{BC}p_{CD}p_{DB} > 0$, $p_{CC}^{(3)} \geq p_{CD}p_{DB}p_{BC} > 0$, and $p_{DD}^{(3)} \geq p_{DB}p_{BC}p_{CD} > 0$ and $p_{AA}^{(5)} \geq p_{AB}p_{BB}^{(3)}p_{BA} > 0$ and $p_{EE}^{(5)} \geq p_{ED}p_{DD}^{(3)}p_{DE} > 0$.

We note that if $p_{ii}^{(n)} > 0$ and $p_{ii}^{(m)} > 0$, then $p_{ii}^{(m+n)} \geq p_{ii}^{(n)} p_{ii}^{(m)} > 0$. Thus, repeating this argument, we have $p_{ii}^{(nk)} \geq (p_{ii}^{(n)})^k > 0$.

This proves that for any even $n = 2k \geq 0$, $p_{ii}^{(2k)} \geq (p_{ii}^{(2)})^k > 0$ for any $i \in \{A, B, C, D, E\}$. Note, if $k = 0$, $p_{ii}^{(0)} = 1 \geq (p_{ii}^{(2)})^0$.

Next, consider n odd with $n \geq 5$. Thus, $n - 3 \geq 2$ and is even. So, we may think $n - 3 = 2k$ for some $k \geq 1$. Thus, for $i \in \{B, C, D\}$, we have $p_{ii}^{(n)} \geq p_{ii}^{(3)} p_{ii}^{(n-3)} = p_{ii}^{(3)} p_{ii}^{(2k)} \geq$

$p_{ii}^{(3)} (p_{ii}^{(2)})^k > 0$. Also, $n - 5 \geq 0$ is even, hence $n - 5 = 2l$ for $l \geq 0$. Thus, for $i \in \{A, E\}$, we have $p_{ii}^{(n)} \geq p_{ii}^{(5)} p_{ii}^{(n-5)} = p_{ii}^{(5)} p_{ii}^{(2l)} \geq p_{ii}^{(5)} (p_{ii}^{(2)})^l > 0$. This proves that $p_{ii}^{(n)} > 0$ for all $n \geq 4$.

2. For any $i, j \in \mathcal{S}$, show that

$$\sup_{n \geq 1} p_{ij}^{(n)} \leq f_{ij}^* \leq \sum_{n=1}^{\infty} p_{ij}^{(n)}.$$

Sample Solution: From the basic renewal equation, for $i, j \in \mathcal{S}$ and $n \geq 1$, we have

$$p_{ij}^{(n)} = \sum_{k=0}^n f_{ij}^{(k)} p_{jj}^{(n-k)} \leq \sum_{k=0}^n f_{ij}^{(k)} \leq \sum_{k=0}^{\infty} f_{ij}^{(k)} = f_{ij}^*$$

using the fact that $p_{jj}^{(n)} \leq 1$ as $p_{jj}^{(n)}$ is the probability of an event. Since this holds for every $n \geq 1$, we have $\sup_{n \geq 1} p_{ij}^{(n)} \leq f_{ij}^*$.

Also, from the basic renewal equation, for $i, j \in \mathcal{S}$ and $n \geq 1$, we have

$$p_{ij}^{(n)} = \sum_{k=0}^n f_{ij}^{(k)} p_{jj}^{(n-k)} \geq f_{ij}^{(n)} p_{jj}^{(0)} = f_{ij}^{(n)}$$

using the fact that $f_{ij}^{(n)} p_{jj}^{(n-k)} \geq 0$ as both terms being probabilities of events and $p_{jj}^{(0)} = 1$. Now summing over all $n \geq 1$, we have

$$\sum_{n=1}^{\infty} p_{ij}^{(n)} \geq \sum_{n=1}^{\infty} f_{ij}^{(n)} = \sum_{n=0}^{\infty} f_{ij}^{(n)} = f_{ij}^*$$

since $f_{ij}^{(0)} = 0$.

3. Consider the stochastic matrix

$$\mathbf{S} = \frac{1}{2} \begin{matrix} & \begin{matrix} 1 & 2 \end{matrix} \\ \begin{matrix} 1 \\ 2 \end{matrix} & \begin{bmatrix} .3 & .7 \\ .6 & .4 \end{bmatrix} \end{matrix}.$$

Show that $\mathbf{S}$ can't be the square of a transition matrix, i.e., there exists no transition matrix $\mathbf{P}$ such that $\mathbf{P}^2 = \mathbf{S}$.

Sample Solution: Suppose that $\mathbf{S} = \mathbf{P}^2$ for a transition matrix where

$$\mathbf{P} = \frac{1}{2} \begin{matrix} & \begin{matrix} 1 & 2 \end{matrix} \\ \begin{matrix} 1 \\ 2 \end{matrix} & \begin{bmatrix} p_{11} & p_{12} \\ p_{21} & p_{22} \end{bmatrix} \end{matrix}.$$

Note that the eigen values of the $\mathbf{P}$ are 1 and $p_{11} + p_{22} - 1$. Thus, the eigen values of $\mathbf{S}$ must be 1 and $(p_{11} + p_{22} - 1)^2$. Hence the trace of $\mathbf{S}$ is $1 + (p_{11} + p_{22} - 1)^2 \geq 1$. But we note that the trace of the matrix $\mathbf{S}$ is $.7 < 1$. This proves that the given matrix can't be the square of a transition matrix.